Code-Based Hard Problems

– Syndrome Decoding & Quasi-Cyclic Variants –

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I. Introduction

Big picture. The syndrome of a linear code is a simple linear map — and it is both the key to perfect decoding and the source of a hard cryptographic problem.

Details to watch. parity-check matrix, syndrome, algebraic structure, NP-hardness, quasi-cyclic.

I. Introduction

The Same Object, Two Different Questions

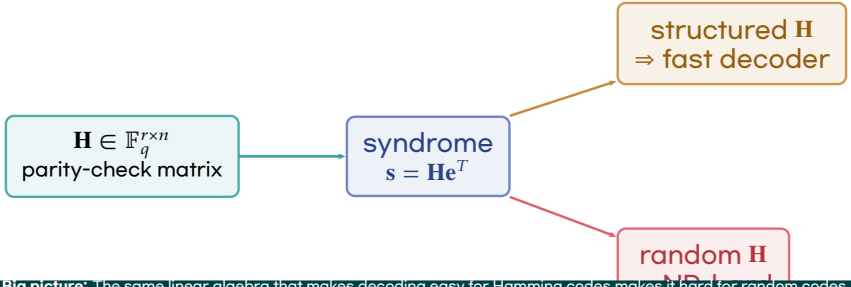


Face 1: Reliable Communication

- A linear code $C \leq \mathbb{F}_q^n$ adds known redundancy.
- Channel adds an unknown error $\mathbf{e}$.
- Syndrome $\mathbf{s} = \mathbf{H}\mathbf{e}^T$ reveals $\mathbf{e}$.
- **Goal**: recover $\mathbf{e}$ quickly — and then correct it.

Face 2: Public-Key Cryptography

- A random $\mathbf{H}$ hides a secret sparse $\mathbf{e}$.
- Adversary sees only $\mathbf{H}$ and $\mathbf{s} = \mathbf{H}\mathbf{e}^T$.
- No algebraic shortcut exists for a random $\mathbf{H}$.
- **Goal**: make recovery computationally infeasible.



Big picture: The same linear algebra that makes decoding easy for Hamming codes makes it hard for random codes.

I. Introduction
The Decoding Gap

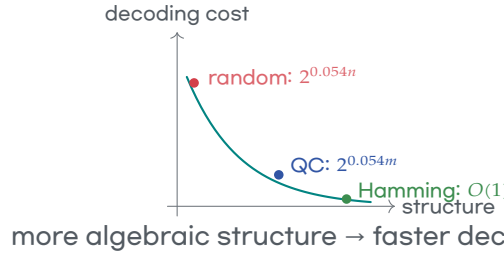


Q. Why does structure matter so much?

Hamming code. For $\mathbf{H}_3 \in \mathbb{F}_2^{3\times 7}$ — columns are all nonzero binary vectors of length 3 — the syndrome $\mathbf{s} = \mathbf{H}_3\mathbf{e}^T$ is the binary address of the error position. Decoding is a single look-up: $O(1)$ time.

Random code. For a random $\mathbf{H} \in \mathbb{F}_2^{r\times n}$, the same equation $\mathbf{H}\mathbf{e}^T = \mathbf{s}$ with $\mathrm{wt}(\mathbf{e}) = w$ defines the **Syndrome Decoding Problem**. The best algorithms run in time $\approx 2^{0.054n}$ — exponential for large n.

What changes? Nothing in the equation. Everything in the matrix structure.



Big picture: The same linear algebra that makes decoding easy for Hamming codes makes it hard for random codes.

I. Introduction
The Central Characters



Code family	Structure	Decoding
Hamming $[2^r-1, 2^r-1-r, 3]_2$	Projective parity checks	Syndrome = error address; $O(1)$
Cyclic (BCH, RS)	Polynomial ideal in R_n	Root syndromes; polynomial-time
QC-MDPC (BIKE)	Circulant blocks in R_m^ℓ	Iterative bit-flipping; probabilistic
Random linear code	None	SDP; best known $2^{0.054n}$

The pattern. More algebraic structure enables a more direct decoder — and makes the code unsuitable as a cryptographic primitive. Random codes have no efficient decoder and are the right hardness engine.

I. Introduction
Roadmap: A Four-Act Story



Act I (here)
Introduction
Two faces of a syndrome

→

Act II
When Decoding
Is Easy
(Hamming, Cyclic)

→

Act III
When Decoding
Is Hard
(SDP, NP-complete)

→

What we build up

- Linear codes, syndromes, cosets
- Hamming: perfect single-error correction
- Cyclic: polynomial-ring structure

Where we arrive

- SDP: NP-complete for random $\mathbf{H}$
- QCSD: structured but still hard
- BIKE and HQC: compact PQC systems

Big picture: The same linear algebra that makes decoding easy for Hamming codes makes it hard for random codes.

II. When Decoding Is Easy

Big picture. Algebraic structure turns the syndrome from a clue into an address: it names the error exactly.

Details to watch. generator matrix, parity-check matrix, syndrome, Hamming bound, perfect code, cyclic ideal, quasi-cyclic hint.

II. When Decoding Is Easy

Linear Codes



Linear $[n,k,d]_q$ Code

A **linear code** is a k-dimensional subspace $C \leq \mathbb{F}_q^n$ with minimum Hamming distance d. Its elements are **codewords**.

Generator matrix

$\mathbf{G} \in \mathbb{F}_q^{k\times n}$: rows form a basis of C. Encoding: $\mathbf{m} \mapsto \mathbf{c} = \mathbf{mG}$.

Parity-check matrix

$\mathbf{H} \in \mathbb{F}_q^{(n-k)\times n}$: $C = \ker(\mathbf{H})$. Check: $\mathbf{c} \in C \Leftrightarrow \mathbf{Hc}^T = \mathbf{0}$.

Parameters

- Rate $R = k/n$ (information density)
- Redundancy $r = n - k$ (check symbols)
- Distance d (error budget)

Error capability

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor \text{ errors corrected}$$

Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

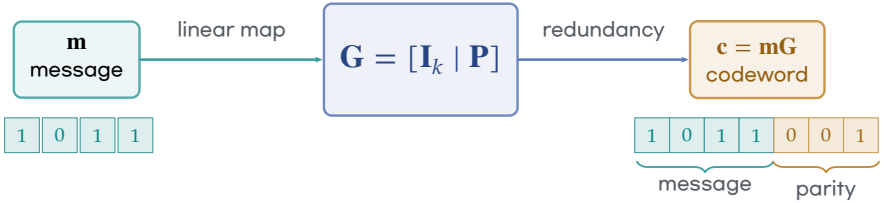
II. When Decoding Is Easy

Systematic Form and Encoding



Systematic form

$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{P}]$: first k coordinates carry the message; last $n-k$ are parity.



Example. For the $[7,4,3]_2$ Hamming code with $\mathbf{m} = (1,0,1,1)$: $\mathbf{c} = \mathbf{mG} = (1,0,1,1,0,1,0)$.

II. When Decoding Is Easy

Syndrome: The Error Fingerprint



Syndrome

For received $\mathbf{y} = \mathbf{c} + \mathbf{e}$,

$$\mathbf{s} = \mathbf{H}\mathbf{y}^T = \underbrace{\mathbf{H}\mathbf{c}^T}_{=\mathbf{0}} + \mathbf{H}\mathbf{e}^T = \mathbf{H}\mathbf{e}^T.$$

The syndrome depends only on the error, not on which codeword was sent.

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\mathbf{H}\mathbf{e}^T} \mathbf{s} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

- $\mathbf{s} = \mathbf{0}$ iff $\mathbf{y} \in C$ (no detectable error).
- Every nonzero $\mathbf{s}$ identifies the coset $\mathbf{e} + C$.
- **Decoding** = find the minimum-weight $\mathbf{e}$ with $\mathbf{H}\mathbf{e}^T = \mathbf{s}$.

Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

Error Correction and the Hamming Bound

Distance theorem

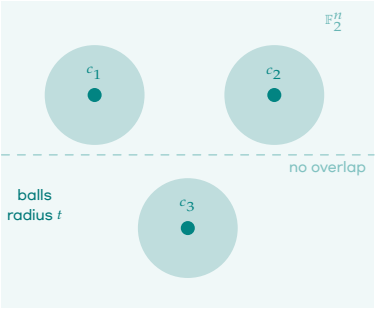
A code of distance d corrects every error of weight $\leq t = \lfloor (d-1)/2 \rfloor$ and detects every error of weight $\leq d - 1$.

Hamming bound

$$q^k \sum_{i=0}^{t} \binom{n}{i} (q-1)^i \leq q^n.$$

Perfect code

Equality in the Hamming bound: correction balls of radius t tile $\mathbb{F}_q^n$ with no gaps and no overlap.



Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

Hamming (1950): The Original Insight



Richard Hamming at Bell Labs in 1947: punched-card overnight jobs crashed on single-bit errors.

His question: "If the machine can detect an error, why can't it fix it?"

Hamming's key idea

Add redundant parity bits so that:

- the syndrome names the **exact error position**, and
- **no syndrome is wasted** — every nonzero syndrome maps to a unique error.

One error at position i, or no error: $n+1$ cases, 2^r syndromes.

$$2^r \geq n+1 \quad \Rightarrow \quad n \leq 2^r - 1$$

Syndrome → action

$000 \rightarrow$ no error
$001 \rightarrow$ flip bit 1
$010 \rightarrow$ flip bit 2
$011 \rightarrow$ flip bit 3
$100 \rightarrow$ flip bit 4
$101 \rightarrow$ flip bit 5
$110 \rightarrow$ flip bit 6
$111 \rightarrow$ flip bit 7

Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

The Binary Hamming Code



Binary Hamming matrix $\mathbf{H}_r$

$\mathbf{H}_r$ is the $r \times (2^r - 1)$ matrix whose columns are all nonzero binary vectors of length r.

$$\mathbf{H}_3 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Binary Hamming code

$\mathrm{Ham}(r, 2) = \ker(\mathbf{H}_r)$, with parameters $[2^r - 1,\ 2^r - 1 - r,\ 3]_2$.

- Every nonzero syndrome identifies a unique column of $\mathbf{H}_r$.
- The column label is the binary address of the error.
- Corrects exactly one error; detects

Perfect

$2^k(1 + n) = 2^{2^r - 1}$ — the correction balls tile all of $\mathbb{F}_2^n$.

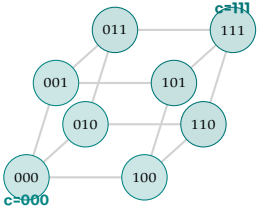
Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

Perfect Tiling: Why Hamming Codes Are Special



For $[3,1,3]_2$ in $\mathbb{F}_2^3$: two correction balls of radius 1 **tile the cube exactly** — no overlap, no gap.



Why "perfect"?

- Each ball: 1 codeword + 3 distance-1 neighbours = 4 words
- Two balls cover all $2^3 = 8$ points: $2 \times 4 = 8$ ☒
- For $\mathrm{Ham}(r,2)$: $2^{2^r-1-r} \times 2^r = 2^{2^r-1}$ ☒

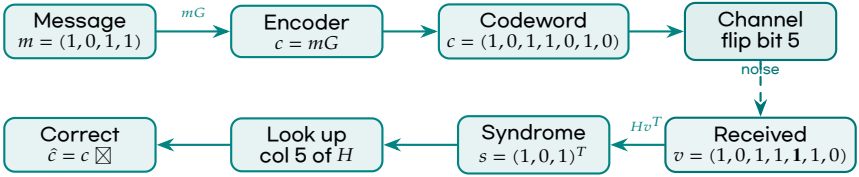
The decoder is one table look-up

Syndrome $\mathbf{s} \neq \mathbf{0}$: find the unique column i of $\mathbf{H}_r$ equal to $\mathbf{s}$. Flip bit i. Time: $O(1)$.

No other linear perfect single-error-correcting code exists (apart from trivial cases).

II. When Decoding Is Easy

Encoding and Correction: The Full Picture



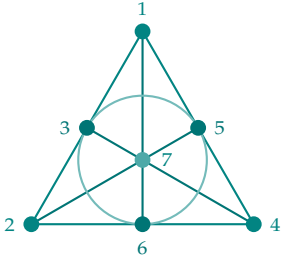
- Syndrome $(1,0,1)^T$ = column 5 of $\mathbf{H}_3$
- Flip bit 5 to recover c ☒
- The decoder never needs to know m

This is why Hamming codes work: the syndrome is literally the binary address of the error. No search, no iteration.

II. When Decoding Is Easy

Geometry Behind the Code: The Fano Plane





The 7 lines ↔ 7 weight-3 codewords:

Line	Columns sum to $\mathbf{0}$
$\{1,2,3\}$	$100 \oplus 010 = 110$
$\{1,4,5\}$	$100 \oplus 001 = 101$
$\{2,4,6\}$	$010 \oplus 001 = 011$
$\{1,6,7\}$	$100 \oplus 011 = 111$
$\{2,5,7\}$	$010 \oplus 101 = 111$
$\{3,4,7\}$	$110 \oplus 001 = 111$
$\{3,5,6\}$	$110 \oplus 101 = 011$

$d = 3$ because the smallest Fano line has 3 points. Algebraic structure = geometric minimum.

Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

Cyclic Codes: Shift Symmetry



Cyclic code

$\mathcal{C} \leq \mathbb{F}_q^n$ is cyclic if $(c_0, \dots, c_{n-1}) \in \mathcal{C} \Rightarrow (c_{n-1}, c_0, \dots, c_{n-2}) \in \mathcal{C}$.

Under the identification $(c_0, \dots, c_{n-1}) \leftrightarrow c(X) = c_0 + c_1 X + \cdots + c_{n-1} X^{n-1}$ in $R_n = \mathbb{F}_q[X]/(X^n - 1)$:

- cyclic shift = multiply by X in R_n
- a cyclic code = an **ideal** in $R_n = \langle g(X) \rangle$ for a unique monic divisor $g(X) \mid X^n - 1$

BCH bound

Prescribe $\delta - 1$ consecutive roots $\alpha^b, \dots, \alpha^{b+\delta-2}$ of $g(X)$. Then $d \geq \delta$.

Hardware bonus

Division by $g(X)$ computes via a shift register with $n - k$ flip-flops. Encoding and syndrome computation run at wire speed.

Example. $X^7 - 1 = (X+1)(X^3 + X + 1)(X^3 + X^2 + 1)$ over $\mathbb{F}_2$. Taking $g = X^3 + X + 1$ gives the cyclic $[7, 4, 3]_2$ Hamming code.

Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

Why Algebraic Structure Makes Decoding Easy



Code family	Structure	Decoding mechanism
Hamming $[2^r-1, *, 3]_2$	Projective parity checks	Syndrome = binary address; $O(1)$ table look-up
Cyclic (BCH, RS)	Ideal in $\mathbb{F}_q[X]/(X^n-1)$	Root syndromes + Berlekamp-Massey; $O(n^2)$
Reed-Solomon	Evaluation code; MDS	Welch-Berlekamp; $O(n^2)$
LDPC / Turbo	Sparse Tanner graph	Belief propagation; near-linear

The pattern. Every efficient decoder exploits a specific algebraic invariant. Projective geometry gives Hamming its $O(1)$ decoder. Polynomial roots give BCH and RS their syndromes.

Remove the algebraic structure → no formula → exponential search.

Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

II. When Decoding Is Easy

Quasi-Cyclic Codes: A Bridge to Cryptography

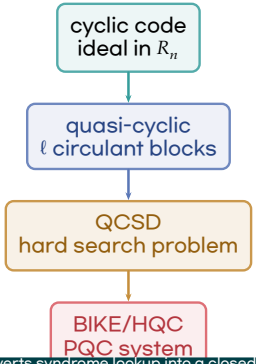


Quasi-cyclic code of index ℓ

$C \subseteq \mathbb{F}_q^n$ $(n = \ell m)$ is **quasi-cyclic** if it is invariant under cyclic shift by ℓ positions. Generator matrices are built from $\ell \times \ell$ **circulant blocks** — each block described by one polynomial in $R_m = \mathbb{F}_q[X]/(X^m - 1)$.

Why QC codes matter in cryptography

- Entire parity-check matrix compressed from $O(n^2)$ to $O(n)$ bits.
- Arithmetic in R_m: polynomial multiplication, inversion.
- Random-looking but algebraically structured.
- Retains hardness of decoding — but now opens new attack surfaces.



Big picture: Algebraic structure — projective geometry, polynomial rings, cyclic symmetry — converts syndrome lookup into a closed-form formula.

III. Syndrome Decoding Problem

Big picture. Remove the algebraic structure of Hamming and cyclic codes, and the same equation $\mathbf{H}\mathbf{e}^T = \mathbf{s}$ becomes NP-complete.

Details to watch. random parity-check matrix, weight constraint, NP-completeness, ISD, McEliece, key size.

III. Syndrome Decoding Problem

The Turn: From Structure to Randomness

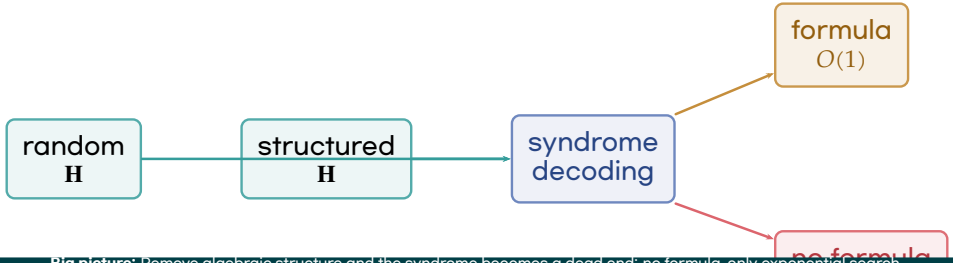


Structured $\mathbf{H}$ (Act II)

- Columns are all nonzero binary vectors (Hamming).
- Generator polynomial of a cyclic ideal (BCH, RS).
- Syndrome points to the error address directly.
- Decoder runs in $O(1)$ or polynomial time.

Random $\mathbf{H}$ (Act III)

- Each entry of $\mathbf{H}$ is an independent uniform bit.
- No projective, polynomial, or cyclic structure.
- Syndrome gives no algebraic hint about $\mathbf{e}$.
- Best decoder takes exponential time.



Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

III. Syndrome Decoding Problem

The Syndrome Decoding Problem (SDP)



Syndrome Decoding Problem

Input: $\mathbf{H} \in \mathbb{F}_q^{r \times n}$, a syndrome $\mathbf{s} \in \mathbb{F}_q^r$, a weight bound $w \in \mathbb{N}$.

Search: find $\mathbf{e} \in \mathbb{F}_q^n$ with

$$\mathbf{H}\mathbf{e}^T = \mathbf{s} \quad \text{and} \quad \mathrm{wt}(\mathbf{e}) \leq w.$$

Decision: does such $\mathbf{e}$ exist?

Parameters

- n: code length
- $r = n - k$: redundancy (rows of $\mathbf{H}$)
- w: target error weight
- $q = 2$: the binary case is most studied

Why hard?

- $\binom{n}{w}(q-1)^w$ candidates — exponential in w
- No algebraic shortcut for random $\mathbf{H}$
- For $n = 1024$, $w = 50$: $\binom{1024}{50} > 10^{80}$

Comparison. For the $[7, 4, 3]_2$ Hamming code with $w = 1$: only 7 candidates; syndrome names the answer in $O(1)$. For a random $[1024, 512]_2$

Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

III. Syndrome Decoding Problem

NP-Completeness of SDP



Berlekamp-McEliece-van Tilborg (1978)

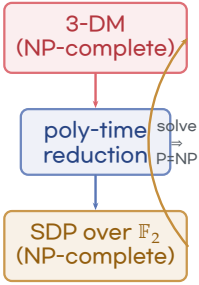
The decision version of the Syndrome Decoding Problem over $\mathbb{F}_2$ is NP-complete.

Proof sketch (reduction from 3-Dimensional Matching).

- Encode a 3-DM instance as a binary matrix $\mathbf{H}$ and syndrome $\mathbf{s}$.
- A perfect 3-D matching corresponds to a weight-w error with $\mathbf{He}^T = \mathbf{s}$.
- The reduction is polynomial time.

Consequence. No polynomial-time algorithm for SDP exists unless $\mathrm{P} = \mathrm{NP}$.

Caution. NP-completeness is a worst-case statement. Cryptographic security requires hardness on average for random instances — a separate



Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

III. Syndrome Decoding Problem

The Combinatorial Explosion



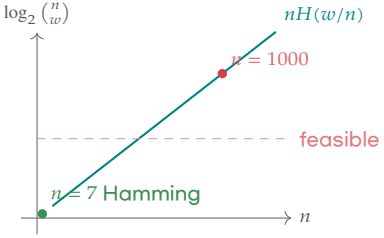
For rate $R = 1/2$ and error weight $w \approx 0.11n$ (near the Gilbert-Varshamov distance), the search space is

$$\binom{n}{w} \approx 2^{H(w/n)\cdot n}$$

where H is the binary entropy function.

At $n = 1000$, $w = 110$: $\binom{1000}{110} \approx 10^{100}$ candidates.

Compare with Hamming. At $n = 7$, $w = 1$: exactly 7 candidates, resolved by the syndrome in $O(1)$.



search space grows exponentially in n

III. Syndrome Decoding Problem

Information-Set Decoding (ISD)



Prange's Algorithm (1962)

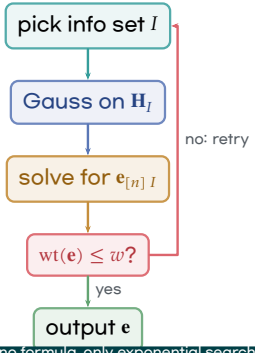
Randomly pick k columns $I \subseteq [n]$ so that $\mathbf{H}_I$ is invertible (an information set). Assume $\mathbf{e}$ is zero on I, solve for $\mathbf{e}_{[n]\,I} = \mathbf{H}_{[n]\,I}^{-1}\mathbf{s}$. If $\mathrm{wt}(\mathbf{e}) \leq w$, output $\mathbf{e}$. Otherwise retry.

Complexity

- Success probability: $p = \binom{n-w}{k} / \binom{n}{k}$
- Cost per trial: $O(n^2)$ (Gaussian elimination)
- Total: $\approx O(n^2/p) = 2^{\Theta(n)}$

Best known ISD variants (binary, rate $1/2$)

- Stern/Dumer (1989): $\approx 2^{0.054n}$ via birthday trick
- BJMM (2012): $\approx 2^{0.049n}$ via representations
- [illegible]: $\approx 2^{0.037n}$



Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

III. Syndrome Decoding Problem

McEliece Cryptosystem (1978)



McEliece public-key encryption

Key generation. Choose an $[n, k, d]$ code $\mathcal{C}$ with a t-error-correcting decoder (e.g., binary Goppa). Let $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ be its generator. Choose random invertible $\mathbf{S} \in \mathbb{F}_2^{k \times k}$, random permutation $\mathbf{P} \in \mathbb{F}_2^{n \times n}$. Publish $\hat{\mathbf{G}} = \mathbf{SGP}$.
Encryption. $\mathbf{c} = \mathbf{m}\hat{\mathbf{G}} + \mathbf{e}$, $\mathbf{e}$ random with $\mathrm{wt}(\mathbf{e}) = t$.
Decryption. Apply $\mathbf{P}^{-1}$, decode $\mathcal{C}$, multiply by $\mathbf{S}^{-1}$.

Security

- Breaking encryption = decoding $\hat{\mathbf{G}}$ without $(\mathbf{S}, \mathbf{P})$ = SDP.
- Binary Goppa codes: efficient algebraic decoder, resistant to structural attacks.
- **Classic McEliece** is a NIST Round 4 finalist.

Classic McEliece (128-bit)

n	3488
k	2720
t	64
Public key	≈ 261 KB
Ciphertext	128 bytes

Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

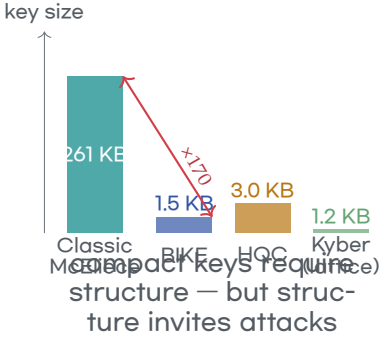
III. Syndrome Decoding Problem

The Price of Generality: Large Keys



Why keys are large

- Security requires $\hat{\mathbf{G}}$ to look random.
- A random $k \times n$ binary matrix has $k(n-k)$ bits.
- For rate $R = 1/2$: $k(n-k) \approx n^2/4$ bits $= O(n^2)$.
- At $n = 3488$: $2720 \times 768 \approx 2.1 \times 10^6$ bits ≈ 261 KB.



The tension

- SDP hardness demands no structure.
- No structure means no compression.
- This is the fundamental trade-off of code-based cryptography.

The question for Act IV: Can we add just enough structure to compress the key while preserving hardness?

Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

III. Syndrome Decoding Problem

SDP: Complexity Landscape



Problem variant	Complexity	Notes
SDP decision	NP-complete	Berlekamp-McEliece-van Tilborg 1978
SDP search, random $\mathbf{H}$, $w \approx \delta_{\mathrm{GV}} n$	$2^{0.054n}$ (classical)	Best: Stern, BJMM, May-Ozerov
SDP search, quantum	$2^{0.027n}$	Grover-accelerated ISD; no exponential improvement
Minimum distance (find min weight codeword)	NP-hard	Related: $\mathbf{s} = \mathbf{0}$ version
Structured SDP (cyclic $\mathbf{H}$)	Unknown; may be easier	Algebraic attacks possible
QC-SDP (quasi-cyclic $\mathbf{H}$)	Believed $2^{\Theta(m)}$	DOOM factor $\sqrt{m}$; Chapter IV

Summary. SDP is the NP-complete engine at the heart of code-based cryptography. Its worst-case hardness is proven; its average-case hardness is conjectured and empirically supported by 45 years of failed attacks. The main weakness: it forces large keys. Act IV resolves this with quasi-cyclic structure.

Big picture: Remove algebraic structure and the syndrome becomes a dead end: no formula, only exponential search.

IV. Quasi-Cyclic Syndrome Decoding

Big picture. Quasi-cyclic structure compresses public keys from megabytes to kilobytes while keeping the essential hardness of SDP — but it opens new attack surfaces.

Details to watch. circulant matrix, QCSD, key compression, BIKE, HQC, DOOM attack, hardness hierarchy.

IV. Quasi-Cyclic Syndrome Decoding

The Engineering Resolution

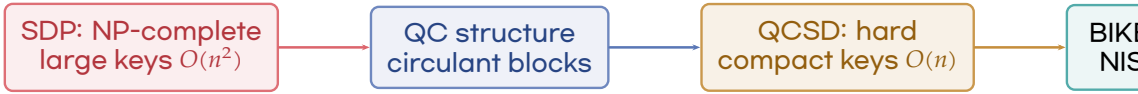


Problem (Act III)

- SDP is NP-complete and quantum-resistant.
- But random $\mathbf{H}$ forces $O(n^2)$-bit public keys.
- Classic McEliece: 261 KB — impractical for many applications.

Resolution (Act IV)

- Add circulant structure: each block described by one polynomial.
- Key size drops to $O(n)$ bits — a factor $\approx n/2$ reduction.
- Hardness inherited from SDP; new attacks studied carefully.



Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

Circulant Matrices: The Building Block

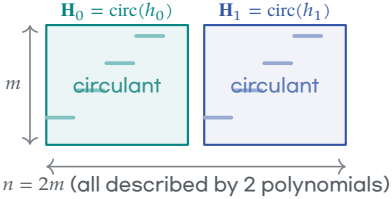


Circulant matrix

$\mathrm{circ}(h_0, \ldots, h_{m-1}) \in \mathbb{F}_q^{m \times m}$: each row is a cyclic shift of the row above.

$$\mathbf{C} = \begin{pmatrix} h_0 & h_1 & \cdots & h_{m-1} \\ h_{m-1} & h_0 & \cdots & h_{m-2} \\ \vdots & & \ddots & \vdots \\ h_1 & h_2 & \cdots & h_0 \end{pmatrix}$$

- $\mathbf{C}$ corresponds to $h(X) = \sum h_i X^i \in R_m = \mathbb{F}_q[X]/(X^m - 1)$.
- $\mathrm{circ}(h) \cdot \mathrm{circ}(g) = \mathrm{circ}(h \cdot g \bmod X^m - 1)$.
- Entire $m \times m$ matrix compressed to m bits.
- Arithmetic in R_m is fast: NTT-based multiplication in $O(m \log m)$.



IV. Quasi-Cyclic Syndrome Decoding

The QCSD Problem



Quasi-Cyclic Syndrome Decoding (QCSD)

Input: block-circulant $\mathbf{H} = [\mathbf{H}_0 \mid \cdots \mid \mathbf{H}_{\ell-1}] \in \mathbb{F}_q^{m \times \ell m}$, $\mathbf{H}_i = \mathrm{circ}(h_i)$, syndrome $\mathbf{s} \in \mathbb{F}_q^m$, weight w.

Search: find $\mathbf{e} = (\mathbf{e}_0 | \cdots | \mathbf{e}_{\ell-1}) \in \mathbb{F}_q^{\ell m}$ with

$$\sum_{i=0}^{\ell-1} \mathbf{H}_i \mathbf{e}_i^T = \mathbf{s} \quad \text{and} \quad \mathrm{wt}(\mathbf{e}) \leq w.$$

Polynomial form: find $e_i \in R_m$ with $\sum_i h_i e_i \equiv s \pmod{X^m - 1}$ and $\sum_i \mathrm{wt}(e_i) \leq w$.

Relation to SDP

- QCSD is a structured instance of SDP.
- A QCSD solver implies an SDP solver (for this instance family).
- QC structure opens new algebraic and statistical attacks.

Why it is still hard

- No polynomial-time algorithm known.
- Algebraic attacks (Gr⊠bner) fail at NIST parameters.
- NIST selected BIKE for standardization consideration.

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

QC-MDPC Codes

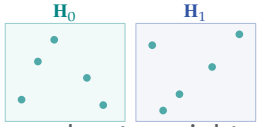


QC-MDPC Code (Misoczki et al., 2013)

A **Quasi-Cyclic Moderate Density Parity Check** code has parity-check matrix $\mathbf{H} = [\mathbf{H}_0|\mathbf{H}_1]$ with $\mathbf{H}_i = \mathrm{circ}(h_i)$ and moderate row weight $d_v \approx \sqrt{n}$.

Key features

- Derived from LDPC philosophy but with higher weight than classical LDPC.
- Efficient probabilistic decoder: **iterative bit-flipping**.
- Security based on QCSD hardness, not on LDPC structure.



Bit-flipping decoder. For each bit, count the parity checks it violates. Flip bits with the highest count. Iterate. This is efficient but probabilistic — decoding failure rate $\delta_{\mathrm{DFR}} > 0$.

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

BIKE: Bit Flipping Key Encapsulation



BIKE KEM (NIST PQC Round 4 finalist)

Setup. QC-MDPC code in R_m^2, m prime, sparse $h_0, h_1 \in R_m$ with $\mathrm{wt} = d_v$.
Key generation. Secret: (h_0, h_1). Public: $g = h_0^{-1} h_1 \in R_m$.
Encapsulation. Sample sparse $e_0, e_1 \in R_m$ with $\mathrm{wt}(e_0) + \mathrm{wt}(e_1) = t$. Ciphertext: $c = e_0 + e_1 \cdot g \in R_m$.
Decapsulation. $c \cdot h_0 = e_0 h_0 + e_1 h_1 = s$ (syndrome). Run BGF bit-flipping decoder on (s, h_0, h_1) to recover (e_0, e_1).

Security

- Breaking = solve QCSD on (h_0, h_1), or find (h_0, h_1) from g.
- Finding (h_0, h_1) from $g = h_0^{-1} h_1$: polynomial factorization with weight constraints in R_m — no polynomial-time algorithm known.

BIKE Level 1 parameters (128-bit)

m	12323
$n = 2m$	24646
d_v	71
t	134
Public key	1541 bytes
Ciphertext	1573 bytes

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

Decoding Failure: A Critical Subtlety

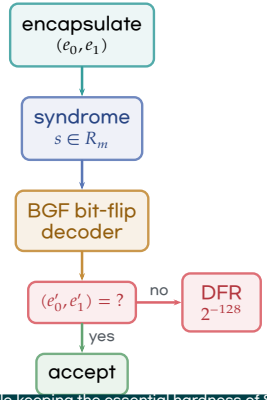
Decoding failure rate (DFR)

The BGF bit-flipping decoder fails with probability δ_{DFR}. For BIKE Level 1: $\delta_{\mathrm{DFR}} \approx 2^{-128}$. A non-zero DFR enables reaction attacks.

The black-peg attack

- Attacker submits chosen ciphertexts and observes decapsulation failures.
- Each failure leaks information about the secret key (h_0, h_1).
- After polynomially many queries, the key can be recovered.

Mitigation. BIKE uses a deterministic BGF decoder variant. The same ciphertext always decapsulates the same way, removing the oracle.



Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

HQC: Hamming Quasi-Cyclic



HQC KEM (NIST PQC Round 4 alternate)

Setup. Work in $R_n = \mathbb{F}_2[X]/(X^n - 1)$, n prime. Choose random $h \in R_n$.
Key generation. Secret: sparse $x, y \in R_n$. Public: $s = x + yh \in R_n$.
Encapsulation. Sample sparse $r_1, r_2, e \in R_n$. Ciphertext: $(\mathbf{u}, \mathbf{v}) = (r_1 + r_2 h,\ mr_2 + sr_1 + e)$.
Decapsulation. $\mathbf{u} \cdot y \approx \mathbf{v} \cdot r_2/r_2$ (inner product extraction), then decode the inner BCH or RS code.

HQC Level 1 (128-bit)

n	17669
Weight of x, y	66
Weight of r_1, r_2, e	75
Public key	2249 bytes
Ciphertext	4481 bytes

Security

- Hardness of QCSD: recover sparse x, y from random-looking $s = x + yh$.
- Uses an inner RS or BCH code for deterministic decryption.
- No DFR issue (deterministic inner decoder).

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

Attack Landscape for QCSD



Attack	Exploits	Notes
ISD (Stern/ BJMM)	No QC structure	Best generic attack; cost $\approx 2^{0.054n}$
DOOM (Sendrier 2011)	QC orbit size m	m cyclically related instances; reduces ISD by $\sqrt{m}$
Algebraic (Gr⊠bner)	Ring R_m	Effective only for tiny parameters; fails at NIST levels
Distinguisher	Statistical QC bias	Can detect QC structure; no decoding from this alone
Reaction attack	DFR > 0	Black-peg attack; mitigated by deterministic BGF decoder
Polynomial factoring	$g = h_0^{-1} h_1$	Recover secret key; no poly-time algorithm known
Quantum ISD	Grover speedup	$\approx 2^{0.027n}$; parameters doubled to compensate

Lesson. QC structure opens more attack surfaces than random SDP. Every known attack is accounted for in BIKE and HQC parameter choices.

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

The DOOM Attack: Cyclic Orbit as Free Speedup



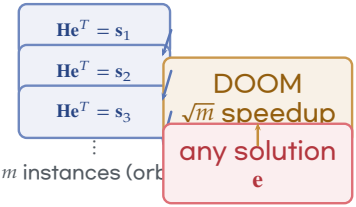
DOOM (Sendrier 2011)

Given N SDP instances with the same $\mathbf{H}$ and different syndromes $\mathbf{s}_1, \ldots, \mathbf{s}_N$, find any one solution. ISD cost: $\approx 2^{0.054n}/\sqrt{N}$.

Connection to QCSD. The cyclic orbit of any solution $\mathbf{e}$ gives m solutions, each with a cyclically shifted syndrome. So QCSD implicitly provides m related instances.

Impact. Effective ISD cost for QCSD: $\approx 2^{0.054n}/\sqrt{m}$. Parameters must be enlarged to absorb this factor $\sqrt{m}$.

Magnitude. For BIKE ($m = 12323$): DOOM gives a speedup of $\sqrt{12323} \approx 2^{6.9}$. Parameters are chosen so that even after this correction, the cost exceeds 2^{128}.

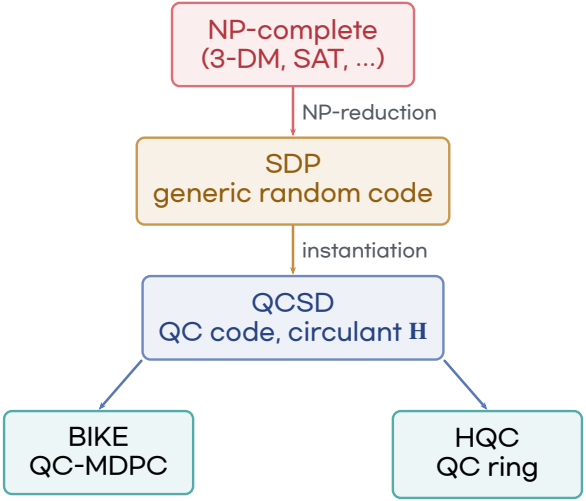


Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

Code-Based Hardness Hierarchy





Reading the hierarchy. Every downward arrow is a reduction: breaking BIKE/HQC implies solving QCSD; solving QCSD implies solving SDP (for this

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

SDP vs. QCSD: The Complete Comparison



Property	SDP (generic)	QCSD
Hardness basis	NP-complete (worst case)	Reduction to SDP; believed hard on average
Public key size	$O(n^2)$ bits	$O(n/\ell)$ bits (circulant compression)
Attack surface	ISD only	ISD + DOOM + algebraic + reaction
Best classical attack	Stern/BJMM $\approx 2^{0.054n}$	ISD with DOOM factor $1/\sqrt{m}$
Best quantum attack	$\approx 2^{0.027n}$	Same + DOOM
Main PQC systems	Classic McEliece (Round 4)	BIKE (Round 4 finalist), HQC (alternate)
Key sizes at 128-bit	≈ 261 KB	1.5-3 KB

The trade-off. QC structure buys a factor $\approx 170\times$ in key size at the cost of a broader attack surface and more careful parameter design.

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

IV. Quasi-Cyclic Syndrome Decoding

Resolution: The Four-Act Story Completed



Act I (Introduction)

The syndrome equation $\mathbf{He}^T = \mathbf{s}$ serves both reliable communication and cryptography. The same math, two opposite goals.

Act II (Development)

Algebraic structure — projective geometry (Hamming), polynomial ideals (cyclic codes) — makes syndrome decoding $O(1)$ or polynomial.

Act III (Turn)

Remove structure and decoding becomes NP-complete (SDP). Hardness is proven. But random codes force $O(n^2)$-bit keys.

Act IV (Conclusion)

Quasi-cyclic structure is the resolution: circulant blocks compress keys to $O(n)$, inherited SDP hardness survives, and careful parameters defeat all known attacks.

Code-based hardness is the art of hiding a sparse

Big picture: Quasi-cyclic structure compresses keys from megabytes to kilobytes while keeping the essential hardness of SDP.

Thank You

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Big picture. Hamming and cyclic codes show that algebraic structure turns syndrome decoding into a closed-form map. Remove that structure and the same syndrome equation becomes NP-hard.

Details to remember. SDP is NP-complete (Berlekamp-McEliece-van Tilborg 1978); QCSD adds quasi-cyclic structure for compact keys; BIKE and HQC are the PQC instances.